

HD74LS74A • Dual D-type Positive Edge-triggered Flip-Flops (with Preset and Clear)

FUNCTION TABLE

Inputs				Outputs	
Preset	Clear	Clock	D	Q	$\bar{Q}$
L	H	X	X	H	L
H	L	X	X	L	H
L	L	X	X	H*	H*
H	H	↑	H	H	L
H	H	↑	L	L	H
H	H	L	X	Q_0	$\bar{Q}_0$

Notes) H; high level, L; low level, X; irrelevant

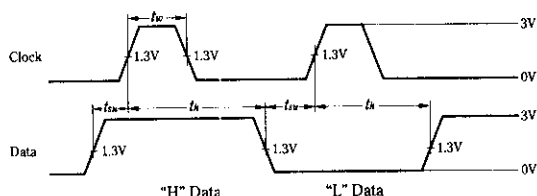
↑; transition from low to high level

Q_0 ; level of Q before the indicated steady-state conditions were established.

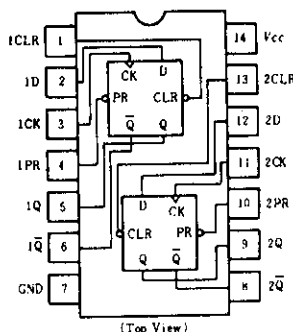
$\bar{Q}_0$; complement of Q_0 or level of $\bar{Q}$ before the indicated steady-state input conditions were established.

*; This configuration is nonstable, that is, it will not persist when preset and clear inputs return to their inactive (high) level.

TIMING DEFINITION



PIN ARRANGEMENT



RECOMMENDED OPERATING CONDITIONS

Item	Symbol	min	typ	max	Unit
Clock frequency	f_{clock}	0	—	25	MHz
Pulse width	Clock High	25	—	—	ns
	Clear/Preset	25	—	—	ns
Setup time	"H" Data	20↑	—	—	ns
	"L" Data	20↑	—	—	ns
Hold time	t_h	5↑	—	—	ns

Note) ↑; The arrow indicates the rising edge.

ELECTRICAL CHARACTERISTICS ($T_a = -20 \sim +75^\circ\text{C}$)

Item		Symbol	Test Conditions	min	typ*	max	Unit
Input voltage		V_{IH}		2.0	—	—	V
		V_{IL}		—	—	0.8	V
Output voltage		V_{OH}	$V_{CC}=4.75\text{V}$, $V_{IH}=2\text{V}$, $V_{IL}=0.8\text{V}$, $I_{OH}=-400\mu\text{A}$	2.7	—	—	V
		V_{OL}	$V_{CC}=4.75\text{V}$, $V_{IL}=0.8\text{V}$, $V_{IH}=2\text{V}$	$I_{OL}=8\text{mA}$ $I_{OL}=4\text{mA}$	—	—	0.5 0.4
Input current	D	I_{IH}	$V_{CC}=5.25\text{V}$, $V_I=2.7\text{V}$	—	—	20	μA
	Clear			—	—	40	
	Preset			—	—	40	
	Clock			—	—	20	
	D	I_{IL}	$V_{CC}=5.25\text{V}$, $V_I=0.4\text{V}$	—	—	-0.4	mA
	Clear			—	—	-0.8	
	Preset			—	—	-0.8	
	Clock			—	—	-0.4	
	D	I_I	$V_{CC}=5.25\text{V}$, $V_I=7\text{V}$	—	—	0.1	mA
	Clear			—	—	0.2	
	Preset			—	—	0.2	
	Clock			—	—	0.1	
Short-circuit output current		I_{OS}	$V_{CC}=5.25\text{V}$	-20	—	-100	mA
Supply current		I_{CC}^{**}	$V_{CC}=5.25\text{V}$	—	4	8	mA
Input clamp voltage		V_{IK}	$V_{CC}=4.75\text{V}$, $I_{IN}=-18\text{mA}$	—	—	-1.5	V

* $V_{CC} = 5\text{V}$, $T_a = 25^\circ\text{C}$

** With all outputs open, I_{CC} is measured with the Q and $\bar{Q}$ outputs high in turn. At the time of measurement, the clock input is grounded.

HD74LS74A

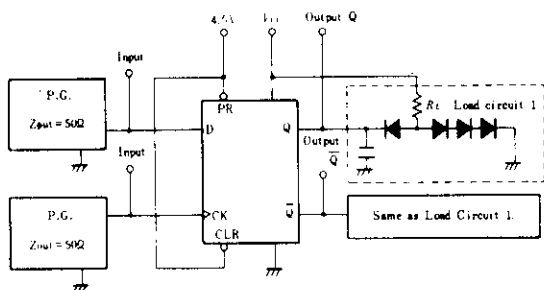
■SWITCHING CHARACTERISTICS ($V_{CC}=5V$, $T_a=25^\circ C$)

Item	Symbol	Inputs	Outputs	Test Condition	min	typ	max	Unit
Maximum clock frequency	f_{max}			$C_L=15pF$, $R_L=2k\Omega$	25	33	—	MHz
Propagation delay time	t_{PLH}	Clock, Clear or Preset	Q, $\bar{Q}$		—	13	25	ns
	t_{PHL}				—	25	40	ns

■TESTING METHOD

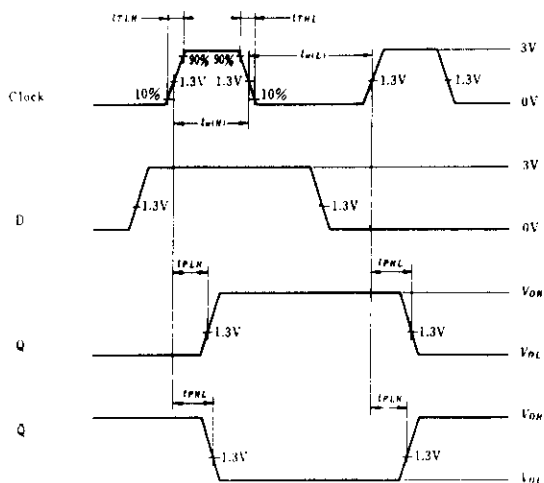
1) Test Circuit

1.1) f_{max} , t_{PLH} , t_{PHL} (Clock→Q, $\bar{Q}$)



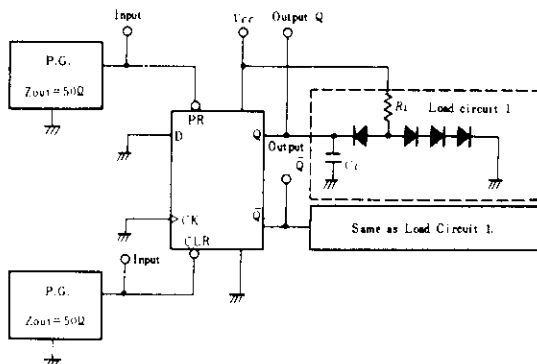
- Notes) 1. Test is put into the each flip-flop
2. All diodes are 1S2074 $\oplus$.
3. C_L includes probe and jig capacitance.

Waveform



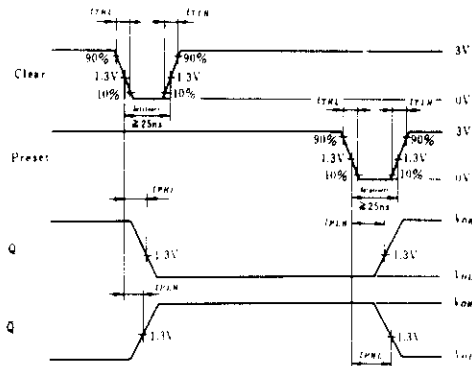
Note) Clock input pulse; $t_{TLH} \leq 15ns$,
 $t_{THL} \leq 6ns$, $PRR=1MHz$, duty
cycle=50% and: for f_{max} ,
 $t_{TLH} = t_{THL} \leq 2.5ns$.

1.2) t_{PHL} , t_{PLH} (Clear or Preset→Q, $\bar{Q}$)

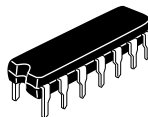
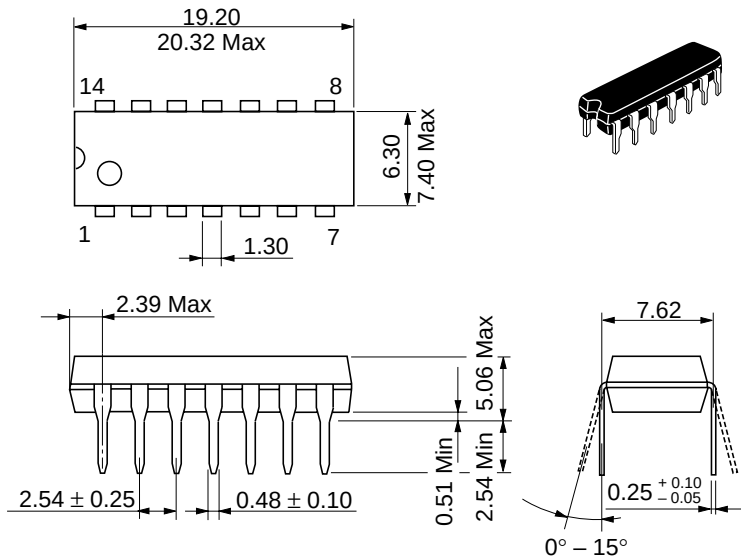


- Notes) 1. Test is put into the each flip-flop
2. All diodes are 1S2074 $\oplus$.
3. C_L includes probe and jig capacitance.

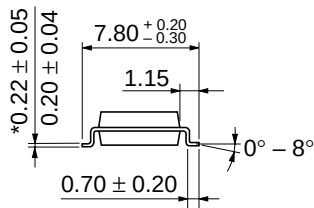
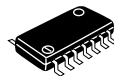
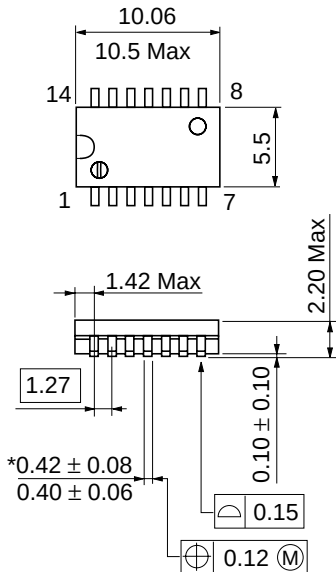
Waveform



Note) Clear and preset input pulse;
 $t_{TLH} \leq 15ns$, $t_{THL} \leq 6ns$,
 $PRR=1MHz$

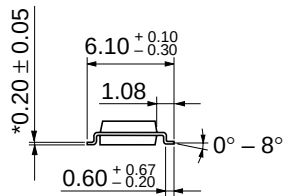
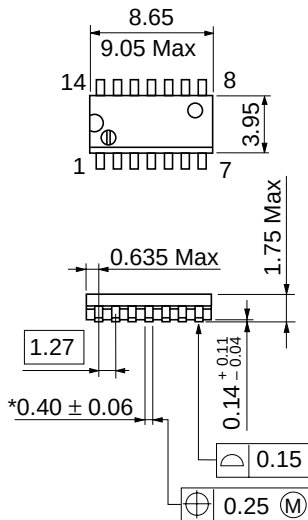


Hitachi Code	DP-14
JEDEC	Conforms
EIAJ	Conforms
Weight (reference value)	0.97 g



Hitachi Code	FP-14DA
JEDEC	—
EIAJ	Conforms
Weight (reference value)	0.23 g

*Dimension including the plating thickness
Base material dimension



Hitachi Code	FP-14DN
JEDEC	Conforms
EIAJ	Conforms
Weight (reference value)	0.13 g

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